

Supreet Gulavani

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Summary

Specialized in RTL design and simulation, bus protocols (AXI, AHB, PCIe), and FPGA prototyping. Proficient in developing memory controllers, cache design, and modular testbenches. Passionate about SoC architecture and embedded system design with a strong foundation in Computer Architecture and Embedded Systems. Currently an independent researcher focusing on DFT tools.

Skills

Protocols & Architectures: PCIe, UART, I2C, SPI, AHBL, AXI, ARM, DDR, Cache, PIC, x86, RISC-V, MIPS, FPGA (Xilinx, Lattice)

Software: C, SystemVerilog, UVM, C++, Python, Bash, Git, Makefile, GDB, Linux, RTOS, edk2, SV Assertions

Hardware: Oscilloscope, Logic Analyzer, PCIe Analyzer, SoC bring-up, Soldering, Hardware Debuggers, Schematic reading

Work Experience

HP (Contract)

May 2024 - Feb 2025

Firmware Engineer

Spring, TX

- Collaborated on M.2 PCIe Gen3 debug tool design and documentation for software interfacing
- Developed a DMA engine in Verilog on Lattice FPGA over PCIe Gen3 X4 via AXI-MM, contributing to board bring-up
- Integrated the ADC IP for real-time on-die junction temperature monitoring; used waveform debugging tools for validation
- Assisted in reviewing schematics for tapeout
- Drafted testing scenarios for ARM-based NPU platforms, contributing to firmware feature development

Intel Corporation

Oct 2022 - Jun 2023

Firmware (BIOS) Developer Intern

Hillsboro, OR

- Ported over 20 test cases to upcoming CPU platforms, enhancing the CI/CD environment for more automated testing
- Resolved Slim Bootloader build script bugs for improved efficiency and reliability
- Created open-source documentation to improve accuracy and reliability, minimizing discrepancies
- Developed structured documentation for internal processes, promoting clarity and organization within the team
- Performed code cleanup and refactoring for Intel's open-source Slim Bootloader, resulting in improved readability
- Authored a whitepaper on Slim Bootloader, showcasing the capabilities and advantages to the team and clients
- Generated maintenance release packages with Yocto, Windows, and UEFI payloads

NXP Semiconductors

June 2022 - Sep 2022

DFT Intern

Austin, TX

- Built an automation tool for ATPG report analysis, improving test coverage assessment
- Worked on SCAN, MBIST, and JTAG flows for subsystem-level regression

Projects

RISC-V 5-stage Pipelined CPU

Oct 2025 - Nov 2025

- Designed a custom RISC-V CPU with a 5-stage pipeline architecture with hazard detection and data forwarding units
- Focused on accuracy, throughput, and modular testbench development by simulating behavior in SystemVerilog using QuestaSim

Verification of RojoBlaze

May 2022 - Jun 2022

- Designed a UVM-style testbench (BFM, scoreboard, monitor, driver) for the 8-bit microcontroller with a 4-stage pipeline
- Achieved 65% functional and 68% code coverage by feeding random constrained test cases through a Makefile

MQTT-based IoT Weather Station - Nexys A7 FPGA

Feb 2023 - Mar 2023

- Built an AXI peripheral-based Ethernet block and interfaced sensors via I2C for temperature, CO2, and TVOC monitoring
- Used LwIP stack with Microblaze CPU softcore to transmit data via MQTT on Standalone OS
- Architected and developed the application control software for the embedded system in C

Space Invaders Arcade Game - Nexys A7 FPGA

Nov 2022 - Dec 2022

- Designed a VGA Controller, Display Timing Generator in SystemVerilog, interfacing via Wishbone for RVFPGA CPU softcore
- Validated the implementation of each peripheral using assembly
- Developed game logic and peripheral drivers in C

DDR4 DRAM Memory Controller

Nov 2021 - Dec 2021

- Simulated command scheduling (in-order, FRFA, out-of-order) with queue logic and timing checks
- Created exhaustive test scenarios to verify DRAM timings and the robustness of the simulator

Education

Portland State University

Sep 2021 - Jun 2023

Master of Science, Electrical and Computer Engineering

Portland, OR

Savitribai Phule Pune University

Aug 2016 - May 2020

Bachelor of Engineering, Electronics and Telecommunication

Pune, India